

Discussion

Finding human-vehicle interactions is a highly semantic task. Describing the person, describing the vehicle, describing an interaction - or at least identifying that there is an interaction and not just proximity - are things that only a VLM can do. Ideally, a VLM would get a video together with a well designed prompt and produce the output by itself. In practice, even for a short clip, a VLM suffers from attention problems and easily misses many interactions.

I started by running a small Qwen model on whole videos locally on my laptop (M4). The result was very poor. Next, I tried Gemini 3.8 Flash via an API. Better, but still not good. Chopping videos into pieces gave the best recall and I ended up with 8-second windows with 4-second overlaps. Also, feeding videos seemed to work better than feeding frames (images).

This windowing resulted in the same interactions being detected multiple times. The descriptions of people, vehicles, and interactions can be slightly different for overlapping windows and it can be hard to determine whether these are the same or different interactions. Ultimately, I assume that we would rather report the same interaction twice than to merge two distinct interactions, and so we have to be careful here.

To give people and vehicles IDs, I have used detection and tracking via Ultralytics. I have used Yolo since it is very light-weight and generally good, even though RF-DETR is possibly better, but it involves registration and licensing issues. Importantly, we use ReID to try to keep IDs stable.

To pass those IDs to a VLM, I just draw the bounding boxes and IDs in the video and then re-encode it. Since it is important to be as non-invasive as possible, I mark only the corners of the bounding boxes and make sure the text is short, thin, and without extra background. Ideally, we would want to pass tracking data separately as text, but I have serious doubts a VLM would understand the combination of ids, coordinates, and timestamps (not tested).

Note that detection and tracking are also imperfect, particularly for blurry low-resolution videos. The problem is mostly with people: they may be unidentified at all or lose their IDs at some point and get new ones later. If a person is not identified in one interaction, we cannot merge that interaction without running a risk of an error. I tried relaxing some parameters; however, it resulted in the ID going to a different person in the proximity of the one who lost it. Textual descriptions from a VLM are also unreliable: it is often confused by front and rear doors and persons' descriptions may be anything (a woman becomes a man, etc.) particularly for blurry videos.

Results

[Output using Gemini API](#)

Gemini API is wildly nondeterministic without any control. I have added support for a locally run VLM; however, open-source VLMs are probably worse while ideally requiring more than a single H-100. The one run on my MacBook was not good enough.

- mKzCQKTHizw_1: Easy.
- mKzCQKTHizw_0: Easy.
- iMGR_0AG3a8_2_3: Blurry video causing detection problems, so we get multiple reports of the same interactions.
- gt1125_06: High resolution, sometimes resulting in hallucinations. Would multiple runs on crops help?
- NmlzoaDcOul_6: Easy.
- NmlzoaDcOul_1: A person goes outside of a frame, then reappears. Also, two people in close proximity. Reidentification issues.
- Hlu4IM4B8hA_1: Blurry video results in the hijacking sometimes not being identified. At least we identify the two as different individuals.
- 1THkHYIQ_bY_0: Slightly challenging. The interaction is non-trivial and we record it as two interactions, which is actually defensible.

Ultimately, to get better results, we would need the strongest models run by ourselves (for determinism). In addition to parameter calibration (taking resolution, fps, etc., into account), we could run the VLM several times, perhaps passing earlier results into prompts (e.g. “is this the same person / interaction?”).

Use of AI

The whole repository is generated by Claude with Codex acting as a reviewer. Additionally, I have actively used ChatGPT to research ideas and libraries. The downsides of this approach include code bloat and a potential for hidden errors. I have tried to simplify and structure the code as much as I could, though the result is not production-grade quality IMO and I would certainly devote more time to simplifications and manual review if I had the time.

On the bright side, this write-up has been 100% done by hand.